

Aleksei Lian

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EDUCATION

York University

Bachelor of Science in Computer Science

Transferred from Carleton University, Ottawa, ON

Toronto, ON

Jan. 2025 – Expected Dec. 2027

Sep. 2023 – Dec. 2024

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Object-Oriented Programming, Theory of Computation

EXPERIENCE

Co-Founder & Lead Engineer

May 2026 – Present

Arzan (Cross-Border Marketplace)

Remote

- Built the product as sole engineer: a **90+** endpoint **FastAPI/PostgreSQL** API, a **React 19/TypeScript** operator console, and an **Expo** mobile app, with **1,600+** backend tests.
- Wrote the **AWS** stack as code in **OpenTofu/Terraform** (ECS Fargate, RDS, ElastiCache, ALB, CloudFront); **GitHub Actions** CI runs every test on SQLite and on PostgreSQL behind PgBouncer.
- Hardened the payment path: captures verified against amount and currency, partial refunds, order expiry, **TOTP** staff 2FA, and an audit log on every privileged mutation.

Co-Founder & Lead Backend Engineer

Jan. 2026 – Present

Tanish (Early-Stage Startup)

Remote

- Shipped a dating platform to **TestFlight**: **FastAPI/PostgreSQL** backend with **2,300+** tests, real-time **WebSocket** chat, and a **React Native** iOS client.
- Designed **JWT** auth with 15-minute access tokens and rotating refresh tokens; replaying an old refresh token revokes its whole chain.
- Integrated **AWS Rekognition** face liveness and a GPT-4o-mini **LLM** answer-alignment score; localized into **4 languages** (English, Russian, Kyrgyz, Kazakh).
- Closed a **privacy leak** on the live database: Supabase's auto-generated Data API exposed user records to anyone with the public key; fixed with deny-by-default **row-level security**.

Math Tutor

Jan. 2024 – Present

Self-Employed

Remote

- Tutor students one-on-one in math: find the exact step where a solution goes wrong and explain it another way.

PROJECTS

Mendy — LLM Advice Companion (Co-Founder) | Python, FastAPI, PostgreSQL, OpenAI API

Jul. 2026 – Aug. 2026

- Built the backend for an **LLM** conversation engine with long-term memory, proactive follow-ups, and tiered safety escalation on **FastAPI**, **async SQLAlchemy 2.0**, and **PostgreSQL** (153 tests).
- Wrote a **40-case safety golden set**; a prompt change ships only at 100% recall on high-risk messages.

Temir AI — Quoting Copilot (Freelance) | JavaScript, Anthropic API, PostgreSQL

Aug. 2026 – Sep. 2026

- Turns a metal order pasted as chat text, PDF, spreadsheet, or photo into a priced quote, invoice, and CSV via the **Anthropic API**, with a deterministic offline parser as fallback.
- Bounded **LLM** spend with per-token cost accounting and per-user, daily, and monthly ceilings serialized by a **Postgres advisory lock**; API key stays server-side; 4 sales managers at 2 warehouse locations used it.

URL Shortener | Python, FastAPI, asyncpg, PostgreSQL, Redis, Docker, GCP

2026

- Engineered an async URL shortener with **FastAPI** and **asyncpg** using connection pooling for concurrent I/O; validated on **100,000 inserts**, each code landing unique.
- CI in **GitHub Actions** (lint plus 38 pytest tests, 23 against a live Postgres container); previously deployed to **Cloud Run** via **Cloud Build**; per-IP **rate limiting** on **Redis**.

Chess Engine (UCI Compatible) | Python

2026

- Implemented **Negamax** with **alpha-beta pruning**, iterative deepening, and **quiescence search**, reaching depth 5–6 in 0.5–1.5s midgame; integrated with Banksia and Arena over the **UCI** protocol.
- Cut redundant search with a **transposition table** (FEN-based hashing) and **MVV-LVA** move ordering, with deadline-based time management to hold UCI time controls.

TECHNICAL SKILLS

Languages: Python, Java, C, TypeScript, JavaScript, SQL

Frameworks: FastAPI, SQLAlchemy 2.0 (async), Pydantic, Alembic, pytest, React, React Native (Expo)

Infrastructure: PostgreSQL, Redis, Docker, AWS (Rekognition), GCP (Cloud Run), OpenTofu/Terraform, Supabase, Vercel, GitHub Actions, Linux

Concepts: REST APIs, WebSockets, Async & Concurrency, Infrastructure as Code, CI/CD, Test-Driven Development, LLM Integration